

Project Name: Talent Tap Pro – Campus Skill-Based Collaboration Engine

1. Overall Approach

To solve the problem of students not finding the right teammates, we designed a **centralized web-based platform** where all collaboration activities happen in one place.

The main idea is **skill-based matching**.

Students create profiles with their skills, and project creators post projects with required roles. The system connects both using matching logic.

The application mainly has two interfaces:

- **Student Dashboard:**
Students can create profiles, view projects, apply to projects, and track their participation.
- **Project Owner Dashboard:**
Users can post projects, define required skills, and select team members.

2. System Architecture Overview

The system follows a simple **3-layer architecture**:

Frontend (Presentation Layer):

- **Student View:**
Allows users to:

- Create profiles with skills
 - View recommended projects
 - Apply to projects
- **Project Owner View:**
Allows users to:
 - Post new projects
 - View applicants
 - Select team members

Backend Logic (Business Layer):

- Acts as the main controller of the system
- Handles:
 - Skill matching logic
 - Project posting and application handling
 - Recommendation system
- Manages workflow like:
Project Posted → Applications Received → Team Selected → Project Started

Data Persistence Layer:

- Stores all important data:
 - User profiles (skills, portfolio)
 - Projects (roles required)
 - Applications
- Ensures data is saved and can be used later

3. Key Components

The system is built using the following main components:

User Profile Management

- Stores user details like:
 - Name
 - Skills
 - Portfolio
- Helps the system understand what the user can do

Skill Matching Engine

- Compares:
 - User skills
 - Project requirements
- Suggests projects to users based on matching skills

Example:

If a student knows Python → system shows Python-related projects

Project Management System

- Allows users to:
 - Post projects
 - Define required roles
 - Manage team members

Application & Selection System

- Students apply to projects
- Project owners review applications
- Team members are selected

Upskilling Support System

- Beginners can join as learners

- Skilled users guide them
- Helps unskilled users become skilled

4. Data Flow Explanation (Talent Tap)

1. **Submission:**

A student logs in, creates a profile with skills, and posts a project or applies to one by selecting required roles (like Designer, Developer, etc.).

2. **Validation:**

The system checks for duplicate applications or project posts. If everything is correct, it saves the data with a timestamp.

3. **Visualization:**

The dashboard updates instantly.

- a. Project owners can see applicants.
- b. Students can see recommended projects based on their skills.
- c. Important or urgent projects can be highlighted.

4. **Resolution:**

The project owner selects team members, updates project status (e.g., “Active”, “Completed”), and progress is tracked in the milestone tracker.

5. Design Decisions (Talent Tap)

• **Web Application:**

Chosen so students can access it easily from laptops or phones without installing anything.

• **Skill Tagging System:**

Skills are categorized (e.g., Coding, Design, Business) to improve matching between students and projects.

• **Priority & Role System:**

Projects include role requirements (High/Medium/Low priority roles) to ensure important positions are filled first.

• **Color-Coded UI:**

Simple visual indicators (like colors or badges) help users quickly identify:

- Urgent projects
- Open roles
- Active vs completed projects

